# Louis Bensard

Co-founder & CTO · AI/ML Engineer
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## SUMMARY

Co-founder & CTO of Sightline and machine learning engineer with 7 years of experience shipping models to production. As the company's first engineer, invented and built its AI recommendation system, the first of its kind in the industry, on top of a transformer forecasting engine trained from scratch in PyTorch. Also built the GPU training stack on AWS, our data platform and the engineering team. Before that, 5 years building and deploying ML models for Fortune 500 clients and leading a team of 9+ engineers.

## EXPERIENCE

**SIGHTLINE** New York, NY
*Co-founder & CTO (previously Founding Head of AI & ML)* *Aug 2025 – Present*
*First engineer. Built the company's AI system, data platform, product and engineering team from scratch, now in production for 10+ enterprise customers.*

- **Forecasting model:** Designed and implemented a PatchTST transformer from scratch in PyTorch (patch tokenization, learned positional and per-series embeddings, multi-head encoder) with a hurdle architecture for zero-inflated demand. Replaced SARIMAX baselines; sub-5% wMAPE across thousands of series.
- **Training and evaluation:** Built the training stack: AdamW with LR scheduling and early stopping, 5-seed training with top-k ensembling, point-in-time backtesting, and an evaluation framework (MAPE, wMAPE, MAE) that gates every model change.
- **Hierarchical forecasting:** Built a second-stage allocation model that disaggregates store-level forecasts to item level, plus the data-preparation pipeline behind training: imputation, cold-start handling and store-migration logic.
- **GPU training platform:** Built model training on AWS Batch: CUDA images in ECR, scheduled retraining via Lambda, versioned model artifacts in S3, inference-only daily runs and experiment tracking in Weights & Biases.
- **Decision intelligence (SCAI):** Invented and built Supply Continuity AI, a recommendation engine new to the category that turns forecasts and inventory projections into scored escalations with a next-best action, recommended order quantities and dates.
- **Data platform:** Designed the event-driven ingestion platform (AWS Transfer Family, S3, SQS, Lambda, ECS, EC2): a config-driven mapper normalizing 7 distributor formats into one canonical schema, idempotent loads into PostgreSQL, and Grafana monitoring with alerting. Millions of rows processed daily.
- **Client implementations:** Technical lead for all our enterprise implementations end to end: scoped data integrations with each client's distributors and IT teams, built the connections and mappings, and managed the lifecycle through go-live.
- **Product and team:** Architected the multi-tenant product (Next.js, TypeScript, Prisma, Vercel) with org isolation and passwordless RDS access via IAM; built the engineering team and standards: hiring, code review, testing, CI/CD and documentation.

  **Tech:** Python, PyTorch, scikit-learn, pandas, SQLAlchemy, PostgreSQL, AWS (Batch, S3, SQS, Lambda, ECS, Transfer Family, RDS, Secrets Manager), Docker, Weights & Biases, Grafana, GitHub Actions, TypeScript, Next.js, Prisma

**EKIMETRICS** New York, NY
*Tech Lead, Machine Learning* *Jul 2024 – Aug 2025*

- **Bayesian modeling:** Built Bayesian marketing-mix models in PyMC with MCMC inference, estimating channel-level effects and ROI from time series KPI data for Fortune 500 retail and beauty clients.
- **Forecasting and deployment:** Built time series forecasting models (PyMC, scikit-learn, XGBoost) and reusable Python/SQL modeling frameworks, taking them from scoping through deployment; $20M+ in measured client impact.
- **Leadership:** Led a team of 9+ engineers; primary technical contact for 10+ strategic accounts, presenting results to C-level executives.

**STRATOS AVIATION** New York, NY
*Founder & Lead AI/ML Engineer* *Aug 2023 – Dec 2024*
*A "Strava for pilots" iOS app, live on the App Store. Designed, built and shipped solo, from concept to launch in 8 months.*

- **Recommender system:** Built personalized airport and route recommendations from nearest-neighbor retrieval over flight time series and geospatial features across 100K airports and 10M+ points of interest, with on-device inference via Core ML.

- **LLM fine-tuning:** Curated the training data and fine-tuned an OpenAI model into a pilot assistant that gives flight-performance recommendations.
- **ML pipelines and security:** Automated preprocessing, feature selection (Boruta, RFE), evaluation and Core ML export; built end-to-end encrypted messaging between pilots (RSA-2048, AES-256).

**IPSOS MMA** New York, NY
*Tech Lead, Machine Learning* *Jul 2019 – Jun 2023*
*Promoted three times from Data Scientist (ML Engineering) to Tech Lead.*

- **Bayesian modeling:** Built Bayesian and hierarchical marketing-mix models (PyMC3, MCMC inference) on time series KPI data for Fortune 500 clients.
- **Predictive modeling at scale:** Developed and deployed hundreds of models, from gradient boosting to neural networks (XGBoost, TensorFlow, scikit-learn), on 10B+ data points; automated the team's feature selection, Bayesian hyperparameter optimization and evaluation pipelines.
- **Interpretability:** Explained model behavior to C-suite executives with SHAP, LIME and interpretable models, on projects worth $4–6M a year.

**PANASONIC AVIONICS CORPORATION** Fullerton, CA
*Machine Learning Engineer (internship)* *Jan 2019 – May 2019*

- Built an in-flight entertainment recommendation engine (scikit-learn, TensorFlow) predicting passenger viewing preferences with 85%+ accuracy on a 10M-row dataset; led a 7-person team.

## EDUCATION

**California State University, Fullerton** 2019
M.S. Statistics, summa cum laude (4.0 GPA, top of class, Phi Kappa Phi). Coursework: Machine Learning, Deep Learning, Reinforcement Learning, Bayesian Analysis, Probability Theory.

**Sorbonne Université, Paris** 2017
B.S. Mathematics and B.S. Computer Science (two degrees), summa cum laude.

## SKILLS

**Deep learning and ML:** Transformers (PatchTST, attention, positional and series embeddings), hurdle and zero-inflated models, multi-seed ensembling, time series forecasting, hierarchical forecasting, recommender systems, Bayesian inference (MCMC), gradient boosting, LLM fine-tuning, model evaluation and backtesting, interpretability (SHAP)

**ML infrastructure:** PyTorch, GPU training (CUDA, AWS Batch), experiment tracking (Weights & Biases), model versioning, training and inference pipelines, Docker, CI/CD (GitHub Actions)

**Data and cloud:** Python, SQL, PostgreSQL, pandas, SQLAlchemy, Alembic, AWS (S3, SQS, Lambda, ECS, Batch, Transfer Family, RDS, IAM, Secrets Manager), Grafana

**Product:** TypeScript, Next.js, React, Prisma, Vercel, Playwright, Swift, Core ML

**Mathematics:** Probability theory, linear algebra, Bayesian statistics, optimization, statistical modeling

## PASSIONS

Licensed private pilot · Competitive rock climber (France, national level) · Overland travel across Africa · English, French (native), Spanish